Fairness Has Memory

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Abstract

When a scarce, indivisible good is allocated by lottery among people with equal claims, the standard theory of fairness says the lottery is the fairest thing available. When the same good is allocated the same way year after year to largely the same people, the theory has nothing further to say: each lottery is fair, so the sequence is fair. This is a mistake. A sequence of individually fair lotteries can be an unfair allocation, and where claimants stay in the pool after being served, fairness requires that their chances in later rounds depend on the history of their unsatisfied claims. The argument uses only premises the standard theory accepts: that a divisible good must be divided in proportion to claims, and that a lottery is a second-best surrogate for division. Over a sequence, the good becomes divisible. I then show what memory should track, in units of the good where later units are as good as earlier ones and in waiting where they are not, and how heavily it may press: only as hard as leaves no claimant's expected share, over whatever stay she has, below what her claim warrants. Where the good recurs, rules that forget and rules that remember lexically are unfair in opposite directions; where winners leave the pool and some will leave unserved, the lottery that forgets is what fairness requires.

1 Two lotteries

Every year Nevada and New Mexico have more people who want to hunt bighorn sheep than sheep to spare, so each holds a lottery. Nevada remembers: an unsuccessful applicant earns a bonus point, her points are squared to fix how many random numbers are drawn for her the next year, and her lowest number is her entry, so ten years of losing gives her a hundred and one draws against a newcomer's one.[1] New Mexico forgets: "New Mexico does NOT grant preference to applicants who were unsuccessful in previous drawings."[2] A hunter passed over for ten years enters the eleventh draw as a first-time entrant.

Many institutions side with Nevada, though they disagree about how: Arizona adds one random number per unsuccessful year; Colorado gives most licences to whoever has the most preference points, chance only breaking ties. Others forget, among them the lotteries that allocate some 85,000 H-1B visas and up to 55,000 Diversity Visas a year. A worker not selected in three successive years enters the fourth draw as a newcomer, even under the rule in force since February 2026, which weights registrations by the wage offered.[3]

The theory of fair allocation has not connected this disagreement to the theory of claims, and where it touches it, sides with New Mexico. The best-developed account, John Broome's (1984a, 1991a, 1991b), holds that fairness requires claims to be satisfied in proportion to their strength, and that when the good is indivisible and claims are equal, an equal lottery is the fairest available procedure. It evaluates a lottery on its own, and on its own each year's

[1] Nevada Department of Wildlife, "Bonus Point Program," https://www.ndow.org/blog/bonus-point-program/.

[2] New Mexico Department of Game and Fish, "How the New Mexico Draw Works," https://wildlife.dgf.nm.gov/hunting/applications-and-draw-information/how-new-mexico-draw-works/.

[3] Department of Homeland Security, "Weighted Selection Process for Registrants and Petitioners Seeking To File Cap-Subject H-1B Petitions," 90 *Federal Register* 60864 (2025).

New Mexico draw is impeccable. Broome's critics, who locate the fairness of lotteries in what they express or exclude rather than in the chances they distribute (Wasserman 1996; Stone 2007, 2011; Spiekermann 2022), reach the same verdict: an impartial procedure is fair whatever last year's did. The wider literature takes the single lottery as its unit (Sher 1980; Kornhauser and Sager 1988; Elster 1988; Duxbury 1999; Saunders 2008; Stone 2008). The nearest to my thesis are two remarks by Hersch and Rowe: that if goods are indexed to times "we lose something that matters for fairness" (2024, 205), and that for recurring chores "who completed the chore last time matters to who has a claim to avoid it this time" (2025, 3). That fairness over a sequence requires remembering it is, by contrast, settled in computational social choice (Lackner 2020; Lackner and Maly 2023; Freeman et al. 2017; Benade et al. 2018); there it is an axiom, here it is derived from the theory of claims, and the derivation fixes how hard memory may press and where it is forbidden (§5).

I will argue for three claims. *Non-separability*: the fairness of a sequence of allocations is not a function of its members' fairness as the standard theory assesses it. *Memory*: when units of a good are allocated repeatedly among claimants who remain in the pool after receiving one, fairness requires that each claimant's chances in later rounds depend on her history of unsatisfied claims. *Currency and weight*: what memory tracks is a claimant's shortfall from her fair share, in units of the good where a later unit is as good to her as an earlier one and in waiting where it is not; and it may press that shortfall only as hard as leaves no claimant's expected share, over whatever stay she turns out to have, below what her claim warrants. Where the good recurs, rules that forget fail the first requirement and rules that let history decide fail the second; where winners leave and some leave unserved, the constraint selects forgetting.

2 Claims, chances, and division

A claim, in Broome's sense, is a duty owed to the candidate herself that she should have the good, as distinct from other reasons for giving it to her, such as usefulness (Broome 1991a, 91–93). Fairness requires that "equal claims require equal satisfaction," that "stronger claims require more satisfaction than weaker ones," and that "weaker claims must not simply be overridden by stronger ones" (Broome 1991a, 95).

Two features of the theory carry my argument. The first I will call the *Division Principle*. Where a good is divisible, proportional satisfaction is achievable and required: if two people have equal claims to some good, it "should be divided," even when giving it all to one of them would do more good (Broome 1991a, 95). Broome states it most generally in a passage that already divides across time:

> The good, for instance, of avoiding military service could be divided by conscripting everyone for a short time rather than a few people for a long time. If people's claims to a good are equal, then fairness requires that the good should be divided between them if it can be. That is the only way to treat them with perfect equality. (Broome 1984a, 47)

The second is the account of lotteries as second best. When the good is indivisible and scarce, a lottery gives "a sort of partial equality in satisfaction": "[E]ach person can be given a sort of surrogate satisfaction. By holding a lottery, each can be given an equal chance of getting the good. This is not a perfect fairness, but it meets the requirement of fairness to some extent" (Broome 1991a, 97–98). Broome grants this only if a chance counts as surrogate satisfaction, which he finds plausible (98). Two things follow. Where division is available, it is required. Where it is not, the surrogate is owed: a claimant who cannot be given her portion must be given her chance.

What duty does Nevada owe anyone to let her shoot a sheep? Broome declines to say which reasons are claims,

beyond granting that some are (Broome 1991a, 93). The claims in my cases arise from the allocator's undertaking: a body that announces it will allocate a public resource fairly among eligible applicants thereby owes each of them fair treatment, and what that requires is not the agency's to stipulate. New Mexico's announcement that it grants no preference to past losers describes its procedure; it does not settle whether the procedure is fair.

3 The divisibility argument

Consider a good that comes in indivisible units, one per period, allocated by one agency among claimants with equal claims, where receiving a unit does not remove a claimant from the pool: a tag a year, a night off the rota a week. Suppose for now that the same N claimants are present for T periods. Within any one period the standard verdict is correct: one unit, N equal claims, no division, so an equal lottery. But the units are indivisible and the sequence is not. Over T periods it is possible to give each claimant T/N units, or as near as the integers allow. Call the procedure that does so *rotation*: draw a random order once and cycle through it. Rotation is, in the sequence, what dividing the food is in the single case. The sequence can be divided, so fairness requires rotation, or whatever comes as close to it as circumstances allow.

Compare New Mexico's procedure, a fresh equal lottery every period: the *forgetting* rule. Over T periods it does not divide the sequence; it raffles it. With ten claimants and ten periods it leaves, on average, three and a half of them with nothing while others receive two or more units.[4] Rotation leaves nobody with nothing. The forgetting rule does over the sequence what the theory forbids in the single case: it takes a divisible good and gives it out by lot.

As a reductio: suppose someone proposed to allocate a cake among ten people with equal claims by cutting it into ten slices and raffling each slice separately. Each raffle, taken by itself, is what the theory recommends for an indivisible good. Nobody thinks this a fair way to allocate a cake; the theory itself says the cake should be divided, and dividing it means giving everyone a slice, not a chance at each slice. A cake handed out over ten days is still a divisible cake.

Two consequences follow. The first is non-separability. The standard theory assesses a lottery by the claims present when it is held, fixed by need, desert and the like, not by what earlier lotteries did. Call a rule *locally fair* at a period if its procedure at that period, so assessed, is fair. The forgetting rule is locally fair at every period; rotation at none after the first, since given the history one claimant is certain to receive the unit. Yet rotation is the fair allocation of the sequence and the forgetting rule is not. So fairness, so assessed, is not separable across occasions. The same result can be put another way, and from §4 on I will put it so: if claims carry history, so that a claimant who has received less than her share has a stronger claim to the next unit, then rotation *is* locally fair at every period and fairness is separable after all. These are one thesis in two vocabularies; what is not available is history-blind claims *and* separable fairness.

The second consequence is memory. Any rule that divides the sequence must give units to those who have received fewer, which requires knowing who has received fewer.

The argument assumes something a careful opponent will press. The Division Principle applies to a good that "could be divided," and the tags a hunter draws in different years are not portions of one tag. What licenses treating them so is *Fungibility*: units allocated in different periods are portions of one divisible good, for a claimant, to the extent that her claim is to a unit of that kind and a later unit would satisfy it as an earlier one would. The hunter's claim is to a tag, not to the 2027 tag. Fungibility is a matter of degree, since delay is nearly always some cost, and it

[4] The expected number with nothing is $N(1-1/N)^T$. The simulation code behind every figure in this paper accompanies it.

fails outright when a later unit is much worse for the claimant: the housing applicant who will be evicted first, the visa applicant whose sponsor will not wait. Then the claim is largely to timeliness. But timeliness is divisible: delay can be shared out as units cannot, so the argument changes currency. Fairness requires that the *waiting* be divided in proportion to claims, where the constraint of §5 allows it (§5.1).[5]

4 The extinction objection

The argument seems to run into a doctrine Broome states in the same year. Once a fair lottery has given Michael the dialysis machine, it would be absurd to hold another on the ground that lotteries are fair: "once Michael has won the first lottery he has a perfect claim to the treatment, and Maggie has none. Consequently it is not at all fair to hold a second lottery" (Broome 1984b, 628–29). Call this the *extinction thesis*: losing a fair lottery extinguishes the loser's claim. Vong (2015) keeps the thesis in one form: a *procedural* claim, once satisfied by a fair lottery, is satisfied, while a *benefit* claim "cannot be lost due to the results of a procedurally fair lottery" (pp. 479–480); so when new units arrive the losers are "a new case of scarce benefit distribution" and get a fresh fair lottery (p. 483). On either form the persistent loser is owed nothing but a fresh lottery among all who remain, which is the forgetting rule; and where the literature does give a lottery memory (Feldblyum Le Blevennec 2023), it is the *winner's* standing that is protected when new claimants arrive.

The reply distinguishes three structures the objection runs together. A *re-run*: the same claimants, the same unit, a second lottery because the first went the wrong way for someone. Broome is right that this is absurd, because there is nothing left to divide. An *interrupted allocation*, Vong's case: one unit, and new claimants arriving before it is handed over; here too there is one unit. And *recurrence*: a new unit each period, among overlapping claimants. Here the extinction thesis is true and irrelevant. Maggie's claim to the 2026 tag was extinguished when she lost; it is someone else's. But there is a 2027 tag, and the Division Principle says the two together are a divisible good, so the 2027 tag goes, or is more likely to go, to someone who did not get the 2026 one. Maggie's claim to the 2027 tag is not the ghost of her claim to the 2026 tag. It is a claim to a unit that exists, with more force than a past winner's for the same reason that, when a cake is divided, someone who has had no slice has more of her claim outstanding than someone who has had one. Where the currency is waiting and everyone will be served, the same holds with waiting in place of slices.

So the extinction thesis and the memory thesis are about different goods. To bite, Broome's objection needs a stronger thesis: that losing a fair lottery for one unit extinguishes any bearing the loss might have on later units. That is what the cake refutes. Someone who raffled each slice could say of each raffle that the losers' claims to that slice were fairly extinguished, and the cake would still not have been divided. Vong's form fails for a related reason: a procedural claim is satisfied by a fresh equal lottery only if a fresh equal lottery is the fair procedure, and over recurring units the Division Principle says it is not. Vong rejects views on which losing weakens a claim and on which winning strengthens it (pp. 477–479); I find no discussion in his paper of the option that losing leaves more of the claim outstanding for *later* units; his objections, which turn on new claimants for a single unit, do not touch it. When new units arrive he treats the losers as "a new case of scarce benefit distribution" (p. 483): the forgetting rule. (For his own case, corneas, where recipients leave the pool, his verdict is right; see §5.1.)

[5] With unequal claims the problem is apportionment: claims problems are apportionment problems (Wintein and Heilmann 2018), and divisor methods have a sequential form that awards seats one at a time by a rank index depending on the tally so far (Balinski and Young 2001, 100, Prop. 3.3). "Repeated" fair division (Igarashi et al. 2024; cf. Aleksandrov and Walsh 2020) concerns envy over many rounds, not claims.

A residual worry makes the surrogate character of chances do real work. If the 2026 lottery gave Maggie surrogate satisfaction, does extra weight in 2027 not satisfy her twice? A chance is not a portion of the good; it is what one gives when a portion cannot be given. If an equal chance of a slice were a kind of equal slice, the raffle of the cake would be a division of the cake, and the theory could not say that the cake should be divided instead. What memory counts is units received, not chances. Wasserman's complaint that claimants cannot eat chances (1996, 47, 49), pressed against calling a chance a partial satisfaction, is, on this view, the beginning of an argument for memory rather than an objection to lotteries.

5 What memory tracks

Sort the institutions by structure. In *recurrence* a claimant who receives a unit stays in the pool: tags, permits, chores. In a *bottleneck* she leaves: a visa, a flat, a kidney. (Hersch and Rowe use "bottleneck" more narrowly, for queues everyone eventually clears.) Structure is fixed by whether winners leave and currency by Fungibility, and in practice they run together: the claimant who leaves on being served was waiting for one unit and cared when it came.

5.1 Shortfall in units

In recurrence the argument of §3 fixes what memory tracks: the difference between the share of units a claimant should have received by now and the number she has. Call this her *shortfall*. With equal claims and one unit per period, a claimant present for r periods who has received w units has shortfall $r/N - w$. Shortfall is neither the number of losses nor time waited: points-for-losses systems reset on a win, so a claimant who has won once in twenty years is treated as a newcomer, and they count a loss among two hundred applicants as a loss among two; time waited tracks shortfall only if units and claimants are constant in number.

Consider the rules that press shortfall with a weight $\kappa \geq 0$: each claimant's chance is proportional to $\max(0,\ 1 + \kappa \cdot \text{shortfall})$. At $\kappa = 0$ this is the forgetting rule. At $\kappa = 1$ a claimant's expected shortfall shrinks by a factor of $1 - 1/N$ each period while no weight is clipped, so it is corrected over roughly the time the practice takes to hand out one round of shares. A past winner's weight falls below a newcomer's, and reaches zero once her surplus reaches $1/\kappa$; she is then excluded until the others catch up, but never taxed, which is the extinction thesis built in. As κ grows the rule approaches giving each unit to whoever is furthest behind, which, when all arrived together, is Aleksandrov et al.'s (2015) Balanced Like, a lottery among those who have received fewest. In a fixed population everyone starts level, so each has the same chance of every unit whatever κ is, and dispersion falls as κ rises. With ten claimants and ten periods the expected number left with nothing falls from 3.5 under the forgetting rule to 2.1 at $\kappa = 1$, 1.4 at $\kappa = 2$ and 0 at $\kappa = 10$. In a fixed population the Division Principle says: press as hard as you can, and rotate.

The rule has near relatives. Lackner's (2020) perpetual voting rules weight voters by the history of past decisions, since "only due to this dependency we can achieve fairness over time" (2104); his Perpetual Quota weights each by $\max(0,\ \text{quota} - \text{satisfaction})$ (2106), which in my setting is shortfall. But his rules are deterministic, so here Perpetual Quota *is* rotation in its tie-breaking order, and the memoryless lottery enters only as a benchmark that "converges to proportional outcomes" but is "not a suitable alternative" over short sequences (2109). The lottery with memory lies between, and the theory of claims requires it, since an arbitrary fixed order among equal claims is what it forbids. Frege's election law, in which votes for unelected candidates carry over while winners pay a cost, is the oldest relative, proportional in the long run with a fixed electorate and not necessarily with a growing one (Harrenstein et al. 2022). Two results bear on the argument. Rules weighting by losses alone fail even simple proportionality (Lackner 2020, Prop. 1; Lackner and Maly 2023, Thm. 2), a deterministic analogue of the failure

of points-for-losses. And when every unit is worth the same to everyone, as here, the forgetting rule's envy between two claimants is a random walk that grows as $\sqrt{T/N}$, while rotation holds it below one unit; the result that no rule beats random allocation in the worst case (Benade et al. 2018, Thms. 2.1, 2.13) needs an adversary who values a unit differently for different claimants, which is where Fungibility fails and the argument of §3 stops. None of this settles how heavily memory may press when claimants come and go, which Lackner leaves open (2109).

5.2 Newcomers and the surrogate constraint

Once claimants come and go, a real objection appears. In the year she first applies, a newcomer has as much of her claim outstanding as any past loser had on her own first application, and may be gone before the division reaches her; under any rule with memory she is given a smaller chance than that loser now has.

The answer also fixes how heavily memory may press. Division of the sequence is available only to a claimant who stays long enough for it to be completed. A claimant who leaves after three periods cannot be given three-tenths of a tag; for her the sequence is as indivisible as the unit. If the allocator knew who would leave, it would owe such claimants the surrogate outright: a chance, each period, proportional to their claim. Not knowing, it owes the surrogate to everyone, in the only form available without knowing tenures in advance: that no claimant's expected share, over whatever tenure she turns out to have, fall short of proportionality to her claims. Call this the *surrogate constraint*. It is measured over the chances the rule assigns, exit taken as given; a claimant who leaves because she lost has a complaint against her luck, not the rule. It is a floor, not a consideration to be weighed against the dispersion that memory reduces, because the two stand differently to the claimants. The chance is what a claimant who will not stay for the division is *owed*: for her there is nothing to divide, and a rule that gives her less partly overrides her claim. Dispersion of shortfall among those who stay is the residue of the lottery that indivisibility made necessary and the floor keeps in place, and a lottery's residue is what the standard theory already *tolerates*; reducing it is an improvement in degree. A rule may not buy an improvement of the second kind with a harm of the first; and since the floor is owed to each claimant who may leave, not to a class, their number does not bear on whether it is owed. Within the floor, those who stay are owed as much division as can be had: the rule that presses shortfall as hard as the surrogate constraint allows.

Suppose claimants leave with probability 0.1 each period, independently of their history, and the pool is kept at ten by replacement (Table 1).

Table 1: Ten claimants with equal claims, one unit per period, each leaving with probability 0.1 per period and replaced; 200 runs of 1,500 periods. A newcomer's chance in her first period (with history-independent exit, also the expected share of those whose tenure turns out to be one period) and expected shares at longer tenures, as multiples of the proportional value $1/N$ per period; standard errors below .005 in the first column, at most .02 elsewhere. Dispersion is the standard deviation, in units, of a claimant's shortfall when she leaves.

Rule	Newcomer's chance	Expected share, tenure 10	Tenure 30	Dispersion of realized shortfall
Forgetting ($\kappa = 0$)	1.00	0.99	0.97	0.95
Shortfall, $\kappa = 1$	1.01	1.00	1.01	0.59

Rule	Newcomer's chance	Expected share, tenure 10	Tenure 30	Dispersion of realized shortfall
Shortfall, $\kappa = 2$	0.97	0.98	1.00	0.49
Shortfall, $\kappa = 10$	0.56	0.94	0.98	0.41
Points for losses, linear (Arizona)	0.26	0.95	1.17	0.70
Points for losses, squared (Nevada)	0.08	0.99	1.20	0.58
Queue / preference points (Colorado)	0.00	1.01	1.26	0.47

Under the forgetting rule expected share is proportional at every tenure, 1.00 by construction; its only defect is the dispersion of realized shortfalls, which is largest. At $\kappa = 1$ the newcomer's chance is proportional to within a per cent, because a past winner's weight reaches zero in only about one claimant-period in twenty; expected share at every tenure is proportional to within the simulation's noise; and dispersion falls by two-fifths. At $\kappa = 2$ the newcomer's chance is 97 per cent of what her claim warrants, a small breach; at $\kappa = 10$ little more than half.

The rules in use breach the constraint outright. Linear points give a newcomer a quarter of the chance her claim warrants and a claimant who stays thirty periods 17 per cent more than her share; squared points cut the newcomer to a twelfth; a queue gives her nothing, and whoever leaves within a few periods leaves with nothing. These rules reduce dispersion, some by more than the moderate shortfall rules, but by transferring expected share from claimants whose tenure turns out short to those whose turns out long, which denies the short-stayers the surrogate they are owed.

So, in recurrence, the third claim is established. Memory is required: the forgetting rule is dominated at no cost in anyone's expected share. But to remember lexically lets an incumbent's accumulated claim override a newcomer's present one. When the pool turns over, fairness requires a rule that raises chances with shortfall at a rate that corrects it over roughly the period in which the practice would ordinarily deliver a share; the region around $\kappa = 1$ has this property, and the rules in use lie well outside it. Such a rule removes some two-fifths of the dispersion the forgetting rule produces; the residual lottery is exactly the surrogate the short-stayers are owed. Only where the pool is fixed is it zero, and there the rule is rotation.

5.3 Bottlenecks and waiting

In a bottleneck nobody accumulates more than one unit; what varies is time waited. Fungibility fails, because each claim is to one unit and to having it soon; what is divisible is the waiting. Under the forgetting rule, waiting is raffled: with a pool of ten kept full by arrivals and one unit per period, about one in seventy waits forty periods or more, though everyone's expected wait is nine. A queue serves everyone at nine. Time, unlike a flat, is divisible, so the Division Principle applies to the waiting and says: equalize it. This is why John and Millum (2020, 181, 205–6) are right that a queue can be as fair as a lottery, and Hersch and Rowe (2024) right to separate "bottleneck" cases, where queues belong, from "scarcity" cases, where lotteries do; between them lies recurrence, where a queue gives a short-stayer none of her accruing share.

The surrogate constraint carries over, and here it cuts the other way. Where claimants may leave unserved, by dying or losing eligibility, and the allocator cannot tell who, those to whom the division is not available are owed the surrogate: an equal chance, each period, at the unit. A pure queue denies it outright; so, to a degree, does any rule that weights by time waited. In recurrence the newcomer's floor could be met because past winners hold a surplus and can be pushed below her; in a bottleneck nobody holds a surplus, since winners leave, so every unit of chance

given to a long waiter is taken from someone whose claim is still wholly outstanding. With exits unforeseeable and uncorrelated with waiting, the only rule under which no tenure class's expected treatment falls below the equal chance is the forgetting rule. So the result is two-sided a second way. Where everyone will be served, the waiting is divisible and fairness requires a queue; where some will leave unserved and the allocator cannot tell who, fairness requires the lottery that forgets. Memory is required in one structure and forbidden in the other; the difference is whether there are surplus-holders to pay for it. Housing lists and transplant lists, where some always leave unserved, fall on the second side: a queue is fairer than a lottery exactly to the extent that exit is foreseeable. (Where an exit is foreseeable and imminent the claim is urgent and the rule must weight it, as transplant allocation does; that is a difference in claims, not a memory of history.)[6]

The verdict for the visa lotteries follows, and it is not what the opening suggests. They are bottlenecks with attrition: a selected worker leaves the pool, and an unselected one may lose her sponsor or status in ways the allocator cannot foresee. So the lottery that forgets is the fair rule, and Hersch and Rowe's (2024, 188) endorsement of the Diversity Visa lottery is right, though not because the case is scarcity rather than bottleneck: the surrogate owed to those who will leave unserved cannot be paid by anyone else. What is wrong with the rule in force since 2026 is not that it forgets but that it weights by wage, a reason of usefulness rather than a claim. And their remark that a periodic lottery in a bottleneck is unfair because a person who "arrives into the lottery pool early and is continuously unlucky" may wait far longer than a lucky newcomer (199) holds only where everyone will be served.

6 How far does memory reach?

If Nevada must remember Maggie's 2026 loss in 2027, must it remember her loss in the housing lottery? In Arizona's draw? Her mother's? A view that cannot stop somewhere principled collapses into a demand that luck be equalized, a different and more contested thesis. Three bounds, besides the bound in weight from §5, follow from the argument that generates the claim.

Memory is bound to the practice. The corrective claim arises because a particular agency's allocation of a particular good left residual unfairness: a claim on the division of *those* units, addressed to *that* agency. The good matters because the 2027 tag can divide the sequence of tags only because it is a tag; a flat is not a portion of what the hunter has a claim to. Goods are individuated by claims, not names: a Nevada tag and an Arizona tag are different goods, because a claim to hunt in one state is not satisfied by a tag for the other, whereas two draws run by one agency for the same tag are one good, so an allocator cannot shed its obligation by splitting a practice in two. The agency matters because dividing is something an agent does with what it has; another allocator has none of its units to give. This distinguishes the thesis from luck egalitarianism (Dworkin 1981; Cohen 1989; Temkin 1993), whose currency is compensation, so that a shortfall in one good can be made up in another; mine concerns the division of one good by the agency dividing it, in that good's currency, and is silent on whether the persistent loser is worse off overall.

Memory is bound to the claim. The corrective claim is a strengthened claim to the good, not a credit. Someone who applied unsuccessfully for ten years and then gave up hunting is owed nothing; there is no tag she has a claim to. Memory is not compensation, since the lottery did no wrong. Two boundary questions follow. *Transition*: when a forgetting practice adopts memory, losses under the old rule count, since the shortfall existed whether or not the rule tracked it; a rectificatory view would say the opposite. *Regress*: the 2028 allocation need not correct the 2027 allocation's failure to correct fully, because shortfall is a state, not a stack of debts; at each period there is one

[6] Dynamic-allocation economics reaches related conclusions by another route (Leshno 2022; Arnosti and Shi 2020), but on grounds of efficiency rather than of what applicants are owed.

quantity to correct.

Memory is bound to the claimant. The claimant is the person, because claims are owed to someone; New Mexico cannot make its persistent losers vanish by declaring that its claimants are "this year's applicants." Nor is there an obligation to correct one's ancestors' shortfalls; the lottery did no wrong.

7 Objections

Losing on purpose. That memory rewards losing is a familiar objection: the NBA moved from a worst-first draft order to a lottery because of the incentive to lose (Elster 1988, 123 n. 46). In the cases that motivate this paper, losing is not under the claimant's control: she can only apply, and applying without a genuine claim is the problem of verifying claims, which every allocation faces. Where losing is controllable, as in a league, fairness is in tension with incentive, a reason to weigh fairness against other things, not evidence that fairness requires forgetting. What must be conceded is that memory raises the value of a manufactured claim: a speculative entrant gains a ticket that grows. The response is to attach memory to participation that costs something, as a paid licence to apply does. That is a constraint on implementing fairness, not evidence that fairness has no memory.

Equality, not fairness. Someone might grant the verdict and deny that fairness is what condemns the outcomes: what is bad about the forgetting rule is inequality, or that some are badly off, matters for a theory of equality or priority (Parfit 2000). Suppose the persistent loser is, all things considered, no worse off than the repeat winner. Her complaint survives: this agency, dividing this good among people with equal claims, has given her less than her share. That complaint is comparative, concerns the good in question, and is addressed to the allocator, the marks of a fairness complaint in Broome's sense; a priority view cannot explain why it lapses when she stops claiming the good.

Ex ante. The last objection comes from the view, descending from Diamond's (1967) reply to Harsanyi (1955) (Frick 2015; Otsuka and Voorhoeve 2009), that a procedure's fairness is a matter of the prospects it gives people at the time of choice. On that view the forgetting rule and rotation are equally fair, since at the outset each claimant has the same prospects under both; the forgetting rule's defect, as the first row of Table 1 showed, is entirely ex post. First reply: the ex ante view must say when the ex ante is. If it is the beginning of the practice, the view cannot prefer the forgetting rule to rotation, since they are ex ante identical, and the Division Principle settles what it leaves open. If it is the beginning of each period, the view asserts separability with respect to history-blind claims, which the cake refutes. Second, the view was developed for one-shot cases, where prospects are all a procedure can distribute. In recurrence what was a prospect becomes a fact the next allocation can attend to.

8 Conclusion

Fairness has memory where the good recurs. What the memory tracks is a claimant's shortfall from her share of the good being divided; it is owed by the agency dividing the good, for as long as she has a claim to it; and it may be pressed only as hard as leaves no claimant's expected share, over whatever stay she has, below what her claim warrants. Where winners leave and some will leave unserved, the same constraint forbids memory, because no one holds a surplus to pay for it. The result was visible in the example Broome chose to illustrate the Division Principle: military service, divided by conscripting everyone for a short time. Everyone's short time is a rotation. The lottery is what one does when one cannot rotate, and one can rotate more often than its defenders have noticed.

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